

习题14.1 p218 1.6.8.10.

习题14.2 p223 7,8,10,15,16.

习题14.3 p228 3道小题, 8和9两题任选一题, 14

习题14.4 p231 1,2,3,4

14.1

14.1.1

Verify $\langle f, g \rangle = \int_a^b fg \, dx$ is an inner product on $C([a, b])$.

Proof: It is clearly a symmetric bilinear form, $\langle f, f \rangle = \int_a^b f^2 \, dx \geq 0$, and equality holds iff $f \equiv 0$.

14.1.6

Prove that all norms on $\mathbb{R}^n$ are equivalent.

Proof: For $\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$ and any norm $\|\cdot\|_2$, $\|\mathbf{x}\|_2 \leq M\|\mathbf{x}\|_1$ where $M = \max_{1 \leq i \leq n} \|e_i\|_2$ is finite. Hence $\|\cdot\|_2$ is continuous under the topology induced by $\|\cdot\|_1$, and it reaches a minimum m on the compact set $\{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\|_1 = 1\}$. Therefore $m\|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2 \leq M\|\mathbf{x}\|_1$ so the two norms are equivalent.

14.1.8

Suppose $A \in \mathbb{R}^{n \times n}$, if $\|A\| < 1$ then $I - A$ is invertible.

Proof: This is the Born expansion $(I - A)^{-1} = \sum_{n=0}^{\infty} A^n$. Furthermore, for $A \in \text{GL}_n(\mathbb{R})$, and $X \in \mathbb{R}^{n \times n}$ with norm small enough, $(A + X)^{-1} = \sum_{n=0}^{\infty} (-1)^n A^{-1}(XA^{-1})^n$.

14.1.10

Consider the norm $\|f\|_{\infty} = \sup_{x \in [0,1]} |f(x)|$ on $C([0, 1])$, is $\|\cdot\|_{\infty}$ induced by some inner product?

Solution: No, since it does not satisfy the parallelogram equation.

14.2

14.2.7

Prove that a sequence $\{\mathbf{a}_m\}$ in $\mathbb{R}^n$ converges iff each $\{\mathbf{a}_m^j\}$ converges.

Proof: Clearly $\max_j |a^j - b^j| \leq |\mathbf{a} - \mathbf{b}| \leq \sum_{j=1}^n |a^j - b^j|$, so using Cauchy criterion $\{\mathbf{a}_m\}$ converges iff each $\{\mathbf{a}_m^j\}$ converges.

14.2.8

Suppose $\{u_i\}, \{v_i\} \in \mathbb{R}^n$ converges to u, v , prove that $\lim_{i \rightarrow \infty} u_i \cdot v_i = u \cdot v$.

Proof: By Ex2.7 each coordinate $\{u_i^j\}$ converges to u^j , hence $\lim_{i \rightarrow \infty} u_i \cdot v_i = \sum_j \lim_{i \rightarrow \infty} u_i^j \cdot v_i^j = u \cdot v$.

14.2.15

Suppose $A \in \mathbb{R}^{n \times n}$ and $\|A\| < 1$, define $B_0 = I_n$ and $B_i = I + AB_{i-1}$, prove that $\{B_i\}$ converges and its limit satisfy $(I - A)B = B(I - A) = I$.

Proof: Recall $I - A$ is invertible. Note that $f : M \mapsto I + AM$ satisfy $\|fM - fN\| \leq \|A\| \cdot \|M - N\|$, so by Banach fixed point theorem B_i converges to the unique fixed point $(I - A)^{-1}$ of f .

14.3

14.3.3

For $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ give the definition of continuity.

Solution: f is continuous iff for any $G \subset \mathbb{R}^m$ open, $f^{-1}(G)$ is open in $\mathbb{R}^n$.

14.3.8

Determine $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y)$ and $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y)$ for $f(x, y) = \frac{(x-y+x^2+y^2)}{x+y}$.

Solution: $\lim_{y \rightarrow 0} f(x, y) = 1 + x$ so $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y) = 1$, and likewise $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y) = -1$.

14.3.14

Prove that $SL_n(\mathbb{R})$ is closed in $\mathbb{R}^{n \times n}$.

Proof: $SL_n(\mathbb{R}) = \det^{-1}(\{1\})$ is clearly closed.

14.4

14.4.1

Prove that for $n \geq 2$, $\mathbb{S}^{n-1}$ is path-connected.

Proof: Stereographic projection gives a homeomorphism between $\mathbb{S}^{n-1}$ and the single-point compactification of $\mathbb{R}^{n-1}$, so $\mathbb{S}^{n-1}$ is path-connected.

14.4.2

Prove that $O(n)$ is compact in $\mathbb{R}^{n \times n}$ but not path-connected.

Proof: Clearly $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$, $M \mapsto MM^T$ is continuous, so $O(n) = f^{-1}(\{I\})$ is closed, and the columns of $A \in O(n)$ are orthonormal implies $O(n)$ is bounded. Hence $O(n)$ is compact in $\mathbb{R}^{n \times n}$.

However, $O(n) \cap \det^{-1}(\{1\})$ and $O(n) \cap \det^{-1}(\{-1\})$ are non-empty closed sets in $O(n)$ so it is not connected, therefore not path-connected.

14.4.3

Suppose $S \subset \mathbb{R}^n$ is both open and closed, prove that $S \neq \emptyset$ or $S = \mathbb{R}^n$.

Proof: $\mathbb{R}^n$ is convex, hence path-connected. If S is non-trivial, take $a \in S$ and $b \notin S$, and consider the path $\gamma : I \rightarrow \mathbb{R}^n$ connecting a, b , then $\gamma^{-1}(S)$ and $\gamma^{-1}(S^C)$ are disjoint open sets that cover I . However $I = [0, 1]$ is connected, a contradiction.

14.4.4

Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is continuous, then f is not injective.

Proof: Restrict f to $f : \mathbb{S}^1 \rightarrow \mathbb{R}$, then suppose $f(1) > f(-1)$, there exists x, y in the two arcs such that $f(x) = f(y) = \frac{f(1)+f(-1)}{2}$, hence f is not injective.